

The University of Texas at Austin
IN-CLASS WORK 9

Introduction to Financial Mathematics

October 01, 2026

EQUITY-LINKED PRODUCTS.

Problem 9.1.

An insurance company sells single premium deferred annuity contracts with return linked to a stock index. The time- t value of one unit of that stock index is denoted by $S(t)$. The contracts offer a minimum guaranteed return rate of g . At time 0, a single premium of amount π is paid by the policyholder, and $\pi \times y$ is deducted by the insurance company. Thus, at the contract maturity date, T , the insurance company will pay the policyholder

$$\pi \times (1 - y) \times \max[S(T)/S(0), (1 + g)^T].$$

You are given the following information:

- The contract will mature in one year.
- The minimum guaranteed rate of return, g , is 2%.
- Dividends are incorporated in the stock index. That is, the stock index is constructed with all stock dividends reinvested.
- $S(0) = 100$.
- The continuously compounded risk-free interest rate is 4%.
- The price of a one-year European call option, with strike price of 102, on the stock index is 12.40.

Determine y so that the insurance company does not make or lose money on this contract.

